B. Fireworks

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: math, number theory | https://codeforces.com/problemset/problem/1945/B

One of the days of the hike coincided with a holiday, so in the evening at the camp, it was decided to arrange a festive fireworks display. For this purpose, the organizers of the hike bought two installations for launching fireworks and a huge number of shells for launching.

Both installations are turned on simultaneously. The first installation launches fireworks every a minutes (i.e., after a, $2 \cdot a$, $3 \cdot a$, ... minutes after launch). The second installation launches fireworks every b minutes (i.e., after b, $2 \cdot b$, $3 \cdot b$, ... minutes after launch).

Each firework is visible in the sky for $m + 1$ minutes after launch, i.e., if a firework was launched after x minutes after the installations were turned on, it will be visible every minute from x to $x + m$, inclusive. If one firework was launched m minutes after another, both fireworks will be visible for one minute.

What is the maximum number of fireworks that could be seen in the sky at the same time?

Input

Each test consists of several test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Then follow the descriptions of the test cases.

The first and only line of each test case contains integers a, b, m ($1 \le a, b, m \le 10^{18}$) — the frequency of launching for the first installation, the second installation, and the time the firework is visible in the sky.

Output

For each set of input data, output a single number — the maximum number of fireworks that can be seen simultaneously.

Example

Input

```
6
6 7 4
3 4 10
7 8 56
5 6 78123459896
1 1 1
1 1 1000000000000000000
```

Output

```
2
7
17
28645268630
4
2000000000000000002
```

Note

In the first set of input data, the fireworks are visible in the sky for 5 minutes. Since the first installation launches fireworks every 6 minutes, and the second one every 7 minutes, two fireworks launched from the same installation will not be visible in the sky at the same time. At the same time, after 7 minutes from the start of the holiday, one firework from the first and one from the second camp will be visible. Thus, it is possible to see no more than 2 fireworks simultaneously.

In the third set of input data, 17 fireworks will be visible after 112 minutes:

- 9 fireworks launched from the first installation at times [56, 63, 70, 77, 84, 91, 98, 105, 112];
- 8 fireworks launched from the second installation at times [56, 64, 72, 80, 88, 96, 104, 112].